

## Personal data

Name: Mohamed Elobaid

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## Research fields

Optimization-based design and control of robotic systems with guarantees

## Current position

**Research Scientist** KAUST, since September 2025.

## Previous positions

**Postdoctoral Researcher**, Istituto Italiano di Tecnologia. Responsible for the joint research initiative with Honda R&D on humanoid robot avatars, and collaborator on the ergoCub new generation of humanoids, February 2022 – August 2025.

**Graduate Teaching Assistant**, Sapienza University of Rome, January 2021 – October 2021.

**Robotics Engineer Fellow**, Istituto Italiano di Tecnologia. Main responsible for the iCub loco-manipulation teleoperation project, January 2018 – November 2018.

**Automation Engineer**, Kenana Engineering and Technical Services. Engineering and commissioning on several new and upgraded sugar factories automation systems, January 2012 – May 2015.

**Teaching Assistant**, University of Khartoum, August 2011 – August 2012.

## Education and titles

**Ph.D. in Automatic Control**, Sapienza University of Rome in *cotutelle* with Université Paris-Saclay, *magna cum laude*. Advisors: Prof. Salvatore Monaco and Dr. Dorothée Normand-Cyrot, November 2018 – February 2022.

**M.Sc. in Control Engineering**, Sapienza University of Rome, 110/110 *cum laude*, September 2015 – October 2017.

**B.Sc. (Hons) in Electrical and Electronic Engineering**, University of Khartoum, second class upper division, November 2006 – August 2011.

## Awards and individual grants

**2024** Best Paper Award, IEEE-RAS Technical Committee on Whole-Body Control, for the preceding year.

**2023** Finalist, IEEE ICRA Outstanding Paper Award on Human-Robot Interaction.

**2019** Best Paper finalist, Intelligent Systems Conference IntelliSys.

**2019** Programme Vinci grant, Université Franco Italienne, grant no. cap. 2 Vinci 2019 C2-897.

## International journal papers

### Published or in press

- J7 C. Sartore, **M. Elobaid**, L. Rapetti, G. Romualdi, S. Daffarra, N. A. Piga, I. Sorrentino, P. M. Viceconte, S. Traversaro, U. Pattacini, L. Fiorio, F. Draicchio, G. Tranfo, L. Natale, M. Maggiali and D. Pucci, “Towards shared embodied intelligence in humanoid robots through optimization, development and testing of the human-aware ergoCub robot,” *Nature Machine Intelligence*, vol. 8, pp. 1221–1237, 2026.
- J6 D. Gorbani, **M. Elobaid**, G. L’Erario, H. A. O. Mohamed and D. Pucci, “Data-fused MPC with guarantees: application to flying humanoid robots,” *IEEE Control Systems Letters*, vol. 9, pp. 2909–2914, 2025.
- J5 **M. Elobaid**, G. Turrise, L. Rapetti, G. Romualdi, S. Daffarra, T. Kawakami, T. Chaki, T. Yoshiike, C. Semini and D. Pucci, “Adaptive non-linear centroidal MPC with stability guarantees for robust locomotion of legged robots,” *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2806–2813, 2025.
- J4 C. Cardenas-Perez, G. Romualdi, **M. Elobaid**, S. Daffarra, G. L’Erario, S. Traversaro, P. Morerio, A. Del Bue and D. Pucci, “XBG: end-to-end imitation learning for autonomous behaviour in human–robot interaction and collaboration,” *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11617–11624, 2024.
- J3 S. Daffarra, U. Pattacini, G. Romualdi, L. Rapetti, R. Grieco, K. Darvish, G. Milani, E. Valli, I. Sorrentino, P. M. Viceconte, A. Scalzo, S. Traversaro, C. Sartore, **M. Elobaid**, N. Guedelha, C. Herron, A. Leonessa, F. Draicchio, G. Metta, M. Maggiali and D. Pucci, “iCub3 avatar system: enabling remote fully immersive embodiment of humanoid robots,” *Science Robotics*, vol. 9, no. 86, eadh3834, 2024.
- J2 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Station-keeping of  $L_2$  halo orbits under sampled-data model predictive control,” *Journal of Guidance, Control, and Dynamics*, vol. 45, no. 7, pp. 1337–1346, 2022.
- J1 **M. Elobaid**, S. Monaco and D. Normand-Cyrot, “Approximate transverse feedback linearization under digital control,” *IEEE Control Systems Letters*, vol. 6, pp. 13–18, 2020.

### Submitted papers

- J8 **M. Elobaid** and E. Feron, “Antipodal initialization for box-constrained quadratic programs,” submitted to *IEEE Transactions on Automatic Control*, April 2026, under review.

### Proceedings of refereed international conferences

- C11 **M. Elobaid**, S. Park and E. Feron, “On the stabilization of rigid formations on regular curves,” IFAC World Congress, 2026, to appear.
- C10 S. Taliani, G. Nava, G. L’Erario, **M. Elobaid**, G. Romualdi and D. Pucci, “Online nonlinear MPC for multimodal locomotion,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2025)*, pp. 12344–12351, 2025.
- C9 **M. Elobaid**, S. Daffarra, E. Ranjbari, G. Romualdi, T. Chaki, T. Kawakami, T. Yoshiike and D. Pucci, “Remote telepresence over large distances via robot avatars: case studies,” in *Proc. IEEE International Conference on Telepresence*, 2024. [TODO: page numbers.]
- C8 L. Rapetti, C. Sartore, **M. Elobaid**, Y. Tirupachuri, F. Draicchio, T. Kawakami, T. Yoshiike and D. Pucci, “A control approach for human–robot ergonomic payload lifting,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2023)*, pp.

7504–7510, 2023. *ICRA Outstanding HRI Paper finalist; IEEE-RAS TC-WBC Best Paper Award 2023.*

- C7 **M. Elobaid**, G. Romualdi, G. Nava, L. Rapetti, H. A. O. Mohamed and D. Pucci, “Online non-linear centroidal MPC for humanoid robot payload carrying with contact-stable force parametrization,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2023)*, pp. 12233–12239, 2023.
- C6 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Virtual holonomic constraints for Euler–Lagrange systems under sampling,” in *Proc. European Control Conference (ECC 2022)*, pp. 693–698, 2022.
- C5 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Digital path-following for a car-like robot,” IFAC Control Conference Africa, 2021, *IFAC-PapersOnLine*, vol. 54, no. 21, pp. 174–179.
- C4 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “On stable right-inversion of non-minimum-phase systems,” in *Proc. IEEE Conference on Decision and Control (CDC 2020)*, pp. 5153–5158, 2020.
- C3 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Sampled-data tracking under model predictive control and multi-rate planning,” IFAC World Congress, 2020, *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 3620–3625.
- C2 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “On unconstrained MPC through multirate sampling,” IFAC Symposium on Nonlinear Control Systems, 2019, *IFAC-PapersOnLine*, vol. 52, no. 16, pp. 388–393.
- C1 **M. Elobaid**, Y. Hu, G. Romualdi, S. Daffarra, J. Babič and D. Pucci, “Telexistence and teleoperation for walking humanoid robots,” in *Proc. Intelligent Systems Conference (IntelliSys 2019)*, pp. 1106–1121, 2019. *Best Paper finalist.*

### Submitted papers

- C13 A. Alshedayfat, A. Abualsaud, A. Al Nasser, O. Wali, A. Laing-Herbert, S. M. Alam, M. O. Tahir, **M. Elobaid** and E. Feron, “Design, control and field validation of a USV–TUV system,” submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).
- C12 Y. Toyomoto, **M. Elobaid**, B. Ferguson, A. Quattrini Li and S. Park, “Safe ergodic control for multi-robot systems via quadratic programming,” submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).

### Ph.D. thesis

*A sampled-data approach in control problems involving partial dynamics cancellation*, advisors Prof. Salvatore Monaco and Dr. Dorothée Normand-Cyrot, Sapienza University of Rome and Université Paris-Saclay, 2022.

### Invited talks and workshops organization

- W1 *Safety-Critical Software and Control for Robotics* workshop co-organization as part of IEEE ARM 2026, Evry, Paris, July 2026.
- P6 *Robust locomotion via online nonlinear MPC*, Honda R&D Engineering, Saitama, Japan, 2023.
- P5 *Feedback linearization via digital control*, tutorial series for researchers at the Istituto Italiano di Tecnologia, Genoa, Italy, 2022.

- P4 *Path-following under digital control*, annual meeting of the Società Italiana Docenti e Ricercatori in Automatica, Ancona, Italy, 2021.
- P3 *Transverse feedback linearization under digital control*, American Control Conference, 2021.
- P2 *On stable right-inversion of partially minimum-phase systems*, IEEE Conference on Decision and Control, 2020.
- P1 *On unconstrained MPC under sampling*, 11th IFAC Symposium on Nonlinear Control Systems, Vienna, Austria, 2019; repeated at the SIDRA annual meeting, 2020.

## Funded research projects

**September 2026 - August 2027** OptiStep: hardware-and-control co-design of an adaptive active ankle-knee prosthesis. KAUST-IIT Joint Call seed project. Proposal conceived and written by M. Elobaid; scientific lead for the KAUST side. Concurrent optimization of hardware parameters and control strategy for a powered, terrain-adaptive lower limb prosthesis.

**February 2022 – February 2025** Honda-IIT joint research on robot avatars. Industrial research project on teleoperation and control of humanoid robots for grasping, positioning and placing packages in warehouse-like environments. Technical lead for IIT.

**January 2024 – June 2024** Software suite for the teleoperation and control of Workfar's Syntro. Technology transfer project delivering a full-stack teleoperation and control software for the Syntro robot. Role: Technical Lead and Project Responsible.

## Further professional and scientific activities

Member of the IEEE Control Systems Society and the IEEE Robotics and Automation Society. Reviewer, over the past seven years, for IEEE Transactions on Robotics, IEEE Robotics and Automation Letters, IEEE Control Systems Letters, and for ACC, CDC, ECC, ICRA, IROS, Humanoids and the IFAC World Congress.

Invited committee member, IFAC Control Conference Africa. 2024.

## Teaching experience

**January 2021 – October 2021** Graduate Teaching Assistant, Sapienza University of Rome. Examination organization for *Nonlinear Systems and Control* (M.Sc.) and *Teoria dei Sistemi* (B.Sc.).

**August 2011 – August 2012** Teaching Assistant, University of Khartoum. Tutorials for *SC5121 Linear Control Theory* and *SC5226 Design of Control Systems*; instructor for *SC4210/SC5124/SC5228 Control Systems Lab I-III*, *EC1102 Computer Systems Lab I*.

**2012, 2015, 2017, 2022** Outreach workshop *Control engineering practice and prospects*, for final-year control and instrumentation students, University of Khartoum.

## Research supervision of Ph.D. students

### Present

**Altayebmohd Alshedayfat**, KAUST, Marine robotics; first results in [C13], since October 2025.

**Ali Abualsaud**, KAUST, Marine robotics; first results in [C13], since October 2025.

**Ali Al Nasser**, KAUST, Marine robotics, first results in [C13], since October 2025.

**Yo Toyomoto**, KAUST, Ergodic Control, first results in [C12], since February 2026.

## Past

**Davide Gorbani**, Istituto Italiano di Tecnologia, supervised his work on data-driven model predictive control in [J6], July 2025 - December 2025.

**Carlos Cardenas-Perez**, Istituto Italiano di Tecnologia, co-supervised his work on end-to-end imitation learning for HRI in [J4], April 2024 - November 2024.

**Ehsan Ranjbari**, Istituto Italiano di Tecnologia, supervised his work on haptic gloves and teleoperation of humanoid robots, and his contribution to the Workfar Syntro project; results in [C9], January 2023 - August 2024.

## B.Sc. and M.Sc. thesis and internship supervision

**Jod Baatiya**, KAUST, internship on explicit inverse kinematics of serial manipulators, June – August 2026.

**Chahine Ghannoudi**, B.Sc. thesis, École Polytechnique de Tunisie, co-supervised his thesis with Prof. Shinkyu Park. Thesis on RL capable simulator for underwater vehicles, defended July 2026.

**Jangkwon Cho**, research internship, KAUST, internship on teleoperation of serial manipulators, January – February 2026.

**Saverio Taliani**, M.Sc. thesis, Istituto Italiano di Tecnologia. Online nonlinear MPC for multimodal locomotion; results in [C10], defended September 2024.

## Open-source software

**walking-teleoperation**. Author of the first release of the teleoperation suite now maintained by the robotology organisation, used for whole-body teleoperation of the iCub and ergoCub humanoids.

**stable-centroidal-mpc**. Reference implementation of the adaptive centroidal MPC of [J5], for bipedal (CasADi/IPOPT) and quadrupedal (ACADOS) platforms.

Contributor to YARP, the **bipedal-locomotion-framework** and **robostack ros-humble**.

## Technical skills

C/C++, MATLAB/Simulink, Python, CasADi, IPOPT, OSQP, CMake, Git, CI/CD, PX4Autopilot, PlatformIO, Pixi, Linux. YARP, ROS and ROS 2, Gazebo, RViz. EC 61131-3, Siemens TIA Portal and PCS7, Yokogawa Centum VP, Wonderware InTouch; P&ID and control architecture design, functional specifications, tendering. Platforms: iCub and ergoCub humanoids, Unitree Aliengo quadruped, Crazyflie aerial vehicles, BlueBoat USV and custom towed underwater vehicle, VR and haptic peripherals.

## Languages

Arabic native; English fluent; Italian basic.

## References

Available upon request.